

Environmental Impact of Networking Devices

Energy Consumption Factors

Networking devices contribute significantly to energy usage through:

- **CPU Processing:** Routing calculations and protocol operations
- **Memory Usage:** Maintaining routing tables and network state
- **Network Traffic:** Packet forwarding and processing load
- **Protocol Overhead:** Update frequency and message processing

*Higher routing activity → More CPU usage → Higher energy consumption
→ Greater environmental impact*

Static vs Dynamic Routing: Environmental Comparison

Parameter	Static Routing	RIP Dynamic Routing
Configuration	Manual, simple	Automatic, complex
CPU Usage	Low	High
Memory Requirements	Minimal	Significant
Update Frequency	None	Every 30 seconds
Network Overhead	Zero	Regular updates
Energy Consumption	Low	High
Environmental Impact	Less	High

Energy-Efficient Routing: Best Practices

Protocol Optimization

- **Adjust Update Intervals:** Increase RIP timers from 30 to 60+ seconds
- **Use Passive Interfaces:** Prevent unnecessary routing updates
- **Implement Route Summarization:** Reduce routing table size
- **Choose Efficient Protocols:** Consider OSPF or EIGRP with better scaling

Hardware Management

- **Port Management:** Disable unused ports and interfaces
- **Link Aggregation:** Consolidate traffic and power down idle links
- **Power Scheduling:** Implement off-peak shutdown procedures
- **Energy-Aware Hardware:** Use devices with power management features